

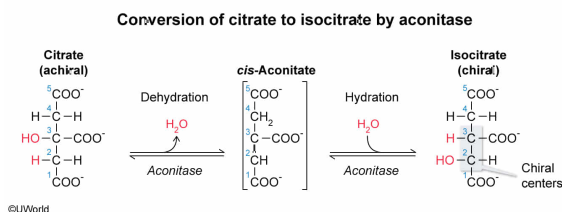
Which of the following correctly describes the stereospecific formation of a citric acid cycle intermediate?

- ☐ A. Citrate synthase consumes ATP to generate citrate from oxaloacetate and acetyl-CoA. [12%]
- ☐ B. Fumarase converts fumarate into both l- and d-malate through a condensation reaction. [23%]
- ☐ C. Succinyl-CoA synthetase consumes GTP to transform succinyl-CoA to succinate. [14%]
- ✔ ☒ D. Aconitase converts citrate into isocitrate through an intermediate. [48%]

Correct

48% answered correctly

Explanation:



Stereoisomers are molecules that contain atoms bonded in the same order but differ in how the atoms are oriented in space. Atoms bonded to four distinct chemical groups are referred to as **chiral centers** or **stereocenters**, which can have different arrangements. **Stereospecific reactions** produce chiral molecules in a specific arrangement. The citric acid cycle produces only two intermediates with chiral centers: D-isocitrate and L-malate. Therefore, only the reactions that form these molecules are classified as stereospecific.

The first reaction in the cycle is the conversion of citrate to isocitrate by the aconitase enzyme through an intermediate called *cis*-aconitate. Aconitase catalyzes consecutive dehydration and hydration reactions that exchange the hydrogen atom and hydroxyl group of the second and third carbon. This isomerase activity results in the transformation of citrate, which is **achiral** (ie, no chiral centers), into isocitrate, which contains **two new chiral centers**.

(Choice A) Synthases such as citrate synthase do not require energy to break or form new bonds, and do *not* consume ATP.

(Choice B) Fumarate is converted to malate by **fumarase** in a stereospecific reaction that generates *only* L-malate; it does *not* produce D-malate. Furthermore, this is a hydration reaction that splits a molecule of water and adds the hydroxyl group and hydrogen atom across the C=C double bond in fumarate; it is *not* a **condensation reaction**.

(Choice C) The energy from cleavage of the **thioester bond** in succinyl-CoA is used by succinyl-CoA synthetase to *produce* GTP, not consume it.

Educational objective:

Chiral centers, also referred to as stereocenters, are atoms bonded to four distinct chemical groups with different possible arrangements in space. Stereospecific reactions produce chiral molecules that have their chiral centers in a specific conformation. The conversion of citrate into isocitrate by aconitase and the transformation of fumarate into malate by fumarase are stereospecific reactions.

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Subject:
Biochemistry

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System:
Metabolic Reactions